

University Physics Quiz 1

Name: _____ SID: _____ School/Department: _____

1. A particle moves outward along a spiral trajectory given by $r = A\theta$, where A is a positive constant. θ increases in time t according to $\theta = \alpha t^2/2$, where α is a positive constant.

(a) Find the time when the radial acceleration of the particle is zero.

(b) Find the time when the radial acceleration and the transverse acceleration of the particle have the equal magnitude.

Solution: From

$$\vec{r} = r\vec{e}_r \quad \dot{\vec{r}} = \dot{r}\vec{e}_r + r\dot{\theta}\vec{e}_\theta \quad \left(\frac{d\vec{e}_r}{dt} = \dot{\theta}\vec{e}_\theta, \frac{d\vec{e}_\theta}{dt} = -\dot{\theta}\vec{e}_r\right) \quad \ddot{\vec{r}} = (\ddot{r} - r\dot{\theta}^2)\vec{e}_r + (2\dot{r}\dot{\theta} + r\ddot{\theta})\vec{e}_\theta,$$

we know that the radial acceleration and the transverse acceleration are respectively

$$a_r = \ddot{r} - r\dot{\theta}^2 \quad \text{and} \quad a_\theta = 2\dot{r}\dot{\theta} + r\ddot{\theta}.$$

Since $\theta = \frac{1}{2}\alpha t^2$ and $r = A\theta = \frac{A}{2}\alpha t^2$, we have

$$\dot{\theta} = \alpha t, \quad \ddot{\theta} = \alpha, \quad \dot{r} = A\alpha t \quad \text{and} \quad \ddot{r} = A\alpha.$$

Substituting above results into the formulae for the radial acceleration and the transverse acceleration, we have

$$a_r = A\alpha - \frac{1}{2}A\alpha^3 t^4, \quad a_\theta = 2A\alpha^2 t^2 + \frac{1}{2}A\alpha^2 t^2.$$

$$(a) \quad a_r = A\alpha - \frac{1}{2}A\alpha^3 t^4 = 0, \quad \alpha^2 t^4 = 2, \quad t = \frac{\sqrt[4]{2}}{\sqrt{\alpha}}.$$

$$(b) \quad a_r = A\alpha - \frac{1}{2}A\alpha^3 t^4 = a_\theta = 2A\alpha^2 t^2 + \frac{1}{2}A\alpha^2 t^2, \quad \alpha^2 t^4 + 5\alpha t^2 - 2 = 0, \quad \alpha t^2 = \frac{\sqrt{33}-5}{2},$$

$$t = \frac{\sqrt{\sqrt{33}-5}}{\sqrt{2\alpha}}.$$

2. A particle travels along the path $y = a + bx + cx^2$ on a plane, where a , b , c are constants. The speed of the particle is constant v_0 .

(a) Determine the x and y components of velocity at $x = 0$.

(b) Find the acceleration of the particle at $x = 0$.

Solution:

$$(a) \quad \dot{y} = b\dot{x} + 2cx\dot{x}.$$

$$\text{At } x = 0, \quad \dot{y} = b\dot{x}.$$

Substituting $\dot{y} = b\dot{x}$ into $\dot{x}^2 + \dot{y}^2 = v_0^2$, we have

$$(1+b^2)\dot{x}^2 = v_0^2.$$

$$\text{Then, we have } v_x = \dot{x} = \frac{v_0}{\sqrt{1+b^2}}, v_y = \dot{y} = \frac{bv_0}{\sqrt{1+b^2}} \quad \text{or} \quad v_x = -\frac{v_0}{\sqrt{1+b^2}}, v_y = -\frac{bv_0}{\sqrt{1+b^2}}.$$

$$(b) \quad \ddot{y} = b\ddot{x} + 2c\dot{x}^2 + 2cx\ddot{x}$$

Making derivative on both side of $\dot{x}^2 + \dot{y}^2 = v_0^2$, we have

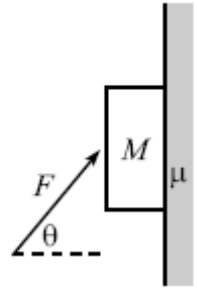
$$2\dot{x}\ddot{x} + 2\dot{y}\ddot{y} = 0, \quad \ddot{y} = -\frac{\dot{x}}{\dot{y}}\ddot{x}$$

$$\text{At } x=0, \quad \ddot{y} = -\frac{1}{b}\ddot{x}, \quad \ddot{y} = b\ddot{x} + 2c\dot{x}^2 \Rightarrow (1+b^2)\ddot{x} = -2bc\dot{x}^2$$

$$a_x = \ddot{x} = \frac{-2bc\dot{x}^2}{(1+b^2)} = \frac{-2bcv_0^2}{(1+b^2)^2},$$

$$a_y = \ddot{y} = -\frac{1}{b}\ddot{x} = \frac{2cv_0^2}{(1+b^2)^2}.$$

3. A book of mass M is positioned against a vertical wall. The coefficient of static friction between the book and the wall is μ_s . You wish to keep the book from falling by pushing on it with a force F applied at an angle θ with respect to the horizontal ($-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$), as shown in figure



(a) For a given θ , what is the minimum F required?

(b) What is the limiting value of θ , below which there does not exist an F that will keep the book up?

Solution: (a)

$$F \sin \theta - Mg + f = 0 \quad (1)$$

$$F \cos \theta - N = 0 \quad (2)$$

$$(2) \Rightarrow N = F \cos \theta$$

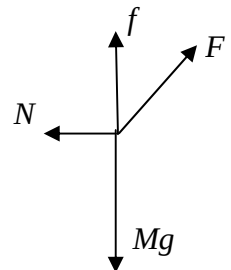
$$f \leq \mu_s N \Rightarrow \mu_s F \cos \theta \geq f \quad (3)$$

$$(1)+(3) \Rightarrow F \sin \theta - Mg + f + \mu_s F \cos \theta \geq f$$

$$F \sin \theta - Mg + \mu_s F \cos \theta \geq 0$$

$$(\sin \theta + \mu_s \cos \theta)F \geq Mg \quad (4)$$

$$F \geq \frac{Mg}{\sin \theta + \mu_s \cos \theta}$$



Free-body diagram

(b) When $(\sin \theta + \mu_s \cos \theta) \leq 0$, the inequality (4) can never be satisfied. This occurs when $\tan \theta \leq -\mu_s$ or

$$\theta \leq \arctan(-\mu_s).$$

4. A child of mass m on a swing raises her center of mass by a small distance b every time the swing passes the vertical position, and lowers her center of mass by the same amount b at each extremal positions (i.e. at the end-point of every swing). The length of the swing (from hanging point to the center of mass of the child) is l .

(a) Assuming small oscillations, calculate the work done by the child per period of oscillation.

(b) Show that the mechanical energy of the swing grows exponentially according to $\frac{dE}{dt} = \alpha E$ and determine the constant α

Solution: (a) Consider a half-period in which the child swing from one extremum to the other. Assume the initial angle to the vertical is θ_i . The center of mass of the child is lowered by b at the extremum and then the child moves to the center/vertical position. The conservation of energy for this part of swing can be written as

$$mg(l+b)(1-\cos\theta_i) = \frac{1}{2}mv_{ci}^2$$

For small oscillations, using the approximation $\cos\theta_i \approx 1 - \frac{1}{2}\theta_i^2$, the above equation can be rewritten as

$$\frac{1}{2}mg(l+b)\theta_i^2 = \frac{1}{2}mv_{ci}^2 \quad (1)$$

where v_{ci} is the velocity when the child first reaches the vertical position. At the vertical position, the child raises her center of mass by b . The angular momentum is conserved during the change. Then, we have

$$mg(l+b)v_{ci} = mg(l-b)v_{cf} \quad (2)$$

where v_{cf} denotes the velocity after the child raises her center of mass at the vertical position. And then child moves to the other extremum and the energy is also conserved for this part:

$$\frac{1}{2}mv_{cf}^2 = \frac{1}{2}mg(l-b)\theta_f^2 \quad (3)$$

where θ_f denotes the angle to the vertical at the other extremum and the small oscillation approximation is also used. Combining (1), (2) and (3) we have

$$\frac{1}{2}mg(l-b)\theta_f^2 = \frac{1}{2}mv_{cf}^2 = \frac{1}{2}m\left(\frac{l+b}{l-b}\right)^2 v_{ci}^2 = \frac{1}{2}mg\frac{(l+b)^3}{(l-b)^2}\theta_i^2.$$

So $\theta_f^2 = \left(\frac{l+b}{l-b}\right)^3 \theta_i^2$. Given $b \ll l$, we can approximate it to $\theta_f \approx \left(1+3\frac{b}{l}\right)\theta_i$ and the $\Delta\theta = \theta_f - \theta_i = 3\frac{b}{l}\theta_i$.

Thus, the energy changed per period (two half-periods) or the work done can be given by

$$\Delta E = 2mgl\left(\frac{\theta_f^2}{2} - \frac{\theta_i^2}{2}\right) = mgl(\theta_f + \theta_i)(\theta_f - \theta_i) \approx 2mgl\theta_i\Delta\theta = 6mgb\theta_i^2 = 12E_i\frac{b}{l}$$

where $E_i = mgl\frac{\theta_i^2}{2}$ is the initial energy.

(b) The change rate of the energy can be expressed as

$$\frac{dE}{dt} \approx \frac{\Delta E}{T}$$

where $T = \frac{2\pi}{\omega}$ and $\omega = \sqrt{\frac{g}{l}}$. Using the result of (a), we have

$$\frac{dE}{dt} = \frac{12\omega b}{2\pi} E = \frac{6\omega b}{\pi} E = \alpha E$$

where $\alpha = \frac{6\omega b}{\pi} = \frac{6bg^{1/2}}{\pi l^{3/2}}$.